

ABSTRACT

The Hostel Management System is a web-based application designed to automate and simplify the management of hostel operations. The system provides an efficient platform for administrators and students to manage various hostel-related activities in a centralized and secure environment. It replaces traditional manual record-keeping methods with a digital solution that improves accuracy, transparency, and operational efficiency.

The system consists of two main modules: Admin and Student. The Admin module enables the administrator to manage courses, rooms, student registrations, attendance, canteen services, complaints, feedback, and report generation. It also provides secure authentication, profile management, password control, and access logging features. The Student module allows users to register, log in, book hostel rooms, view attendance records, order food, submit complaints, provide feedback, and manage their profiles.

The application is developed using PHP for backend processing and MySQL for database management, while HTML, CSS, and JavaScript are used for the user interface. The system follows a modular architecture, ensuring scalability, maintainability, and secure data handling. Proper validation mechanisms and session management techniques are implemented to maintain data integrity and user authentication.

Overall, the Hostel Management System enhances administrative efficiency, reduces paperwork, minimizes human errors, and provides a user-friendly platform for managing hostel activities effectively.

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CHAPTER 1

INTRODUCTION.

1.1 PROJECT DESCRIPTION

The Hostel Management System is a web-based application developed to manage hostel activities in a simple and efficient way. It helps in handling student registration, room allocation, fee management, attendance tracking, food services, and complaint management through a centralized system. The application replaces manual record keeping with a digital platform that stores all data securely in a database.

The system is divided into two main modules: Admin and Student. The Admin module manages rooms, students, fees, attendance, food items, and complaints. The Student module allows students to register, book rooms, view attendance, pay fees, order food, and submit complaints. The system improves accuracy, reduces paperwork, and ensures smooth communication between students and hostel administration.

1.2 PROBLEM STATEMENT

In many institutions, hostel management is handled manually using registers and paper records. This method is time-consuming and may lead to errors, data loss, and difficulty in tracking student and room details. Managing fees, attendance, and complaints manually becomes complicated as the number of students increases.

There is a need for a digital system that automates hostel operations and maintains records securely. The Hostel Management System is developed to reduce manual work, improve efficiency, ensure data accuracy, and provide real-time access to information for both administrators and students.

1.3 EXISTING SYSTEM

In the existing system, hostel management activities are carried out either manually or using basic semi-computerized tools. Student registration is completed through paper-based forms, and room allocation details are maintained in physical registers. Attendance is recorded in manual attendance books, while fee payments are tracked through handwritten records. Complaints are submitted through written applications, and food orders are managed verbally or entered manually. This traditional approach leads to several limitations. It is time-consuming, as maintaining and updating records manually requires significant effort. There is a high possibility of data inaccuracy due to human errors in writing or updating information.

The system also lacks proper security, making sensitive student data vulnerable. Additionally, there are no real-time updates, so students cannot instantly check room availability or complaint status. Generating monthly reports and summaries is difficult and must be done manually, and searching for old records becomes challenging due to poor data management. Based on literature studies and previous systems, it is clear that automation can greatly improve efficiency, transparency, accuracy, and overall management of hostel administration.

1.4 PROPOSED SYSTEM

The proposed system is a web-based Hostel Management System developed to automate hostel administration processes. It replaces the manual method of managing student records, room allocation, attendance, food services, and complaints with a centralized digital platform.

The system consists of two main modules: Admin and Student. The Admin module manages students, rooms, attendance, food services, complaints, feedback, and reports with secure authentication. The Student module allows students to register, log in, book rooms, view attendance, order food, submit complaints, and manage profiles.

The system is developed using PHP and MySQL with HTML, CSS, and JavaScript for the user interface. It improves efficiency, reduces paperwork, minimizes errors, and ensures smooth hostel management.

The proposed Hostel Management System offers several advantages over the existing manual system. It automates all major hostel activities, eliminating paperwork and manual record maintenance by storing student details, room allocation, attendance, and food orders digitally. This improves accuracy because data is stored in a structured database, reducing calculation errors and duplication. The system saves time by allowing quick student registration, room allotment, attendance marking, and report generation.

It provides centralized data management, where all hostel-related information is stored in a single database for easy access and updates. Secure authentication ensures that only authorized admin and students can log in and access data. Complaint handling becomes easier as students can submit complaints online and track their status. The system also improves communication through feedback and complaint modules. It can generate structured reports such as attendance, food, and student reports. Additionally, the system is scalable and can be expanded in the future with features like online payments and SMS notifications. Overall, it offers a user-friendly interface with a simple dashboard for smooth navigation.

CHAPTER 2

SYSTEM DESIGN

2.1 ARCHITECTURE DESIGN

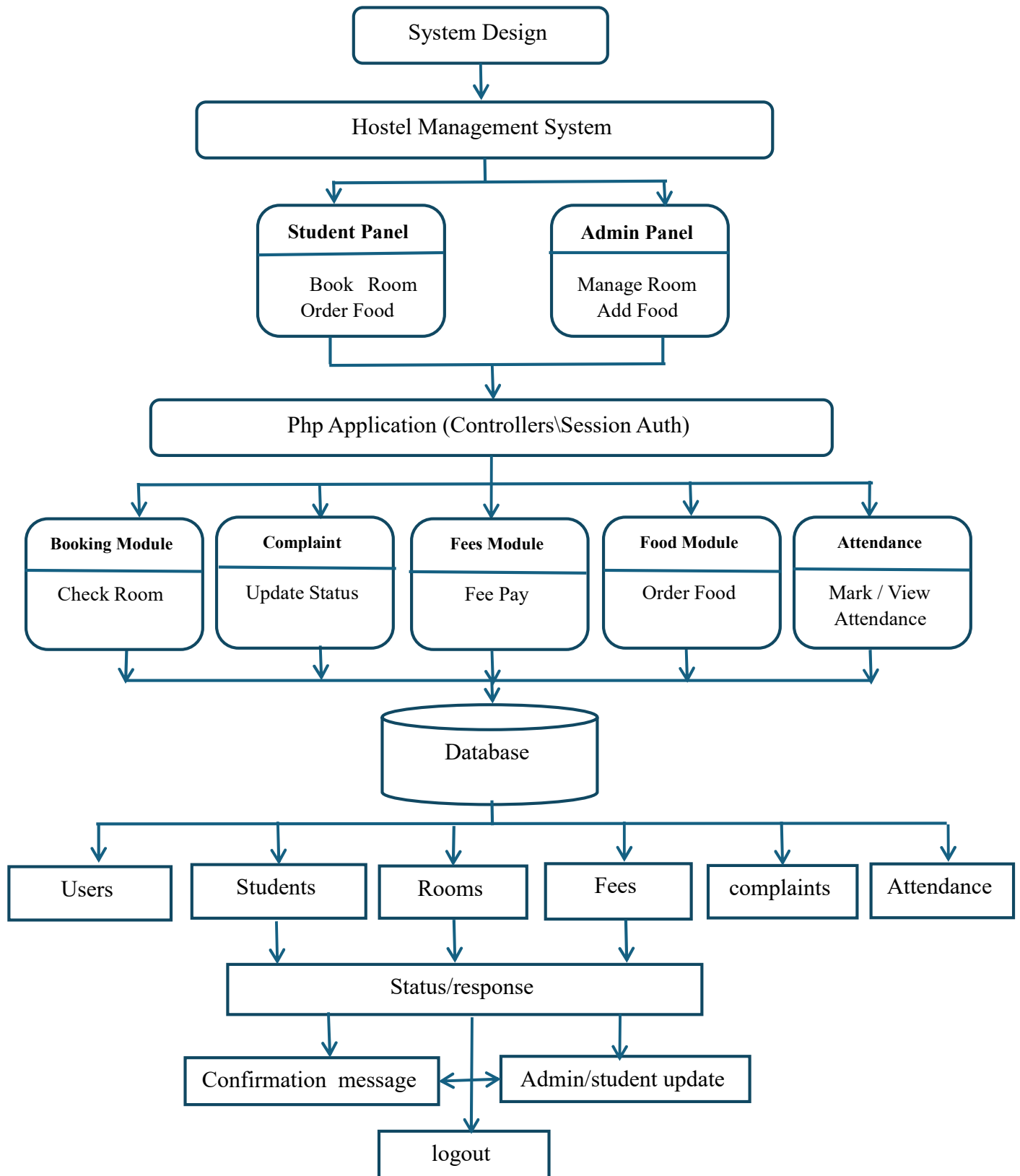
The system is designed as a comprehensive web-based Hostel Management Platform that centralizes and automates all major hostel operations through well-structured and integrated modules for two primary user roles: Admin and Student. The platform is developed to provide an efficient, transparent, and secure environment for managing hostel activities digitally.

The system operates using server-side logic and database-driven processing, where every module performs specific management tasks independently while still being connected to a centralized database. The Admin module enables full control over hostel operations, including managing student records, allocating rooms, monitoring attendance, handling food management, reviewing complaints, and generating reports. The Student module allows students to register, update their profiles, view room details, check attendance, place food orders, submit complaints, and provide feedback.

The student management module handles student registration, profile updates, and record maintenance. The room allocation module manages room availability, allotment, and occupancy status. The attendance tracking module records and monitors daily attendance efficiently. The canteen management module allows students to place food orders while enabling administrators to manage food records. The complaint handling module facilitates online complaint submission and tracking, improving communication between students and hostel authorities.

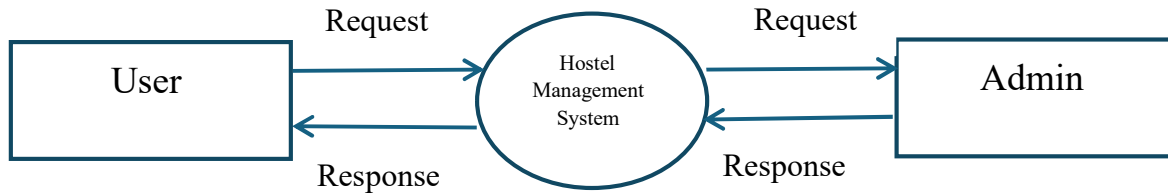
All these modules interact with a centralized database system to securely store, retrieve, and update information in real time. The use of structured database tables ensures data accuracy, consistency, and integrity. Overall, the system provides a reliable, scalable, and user-friendly digital solution that enhances efficiency, reduces manual work, improves transparency, and ensures better management of hostel operations.

2.1.1 Architecture diagram

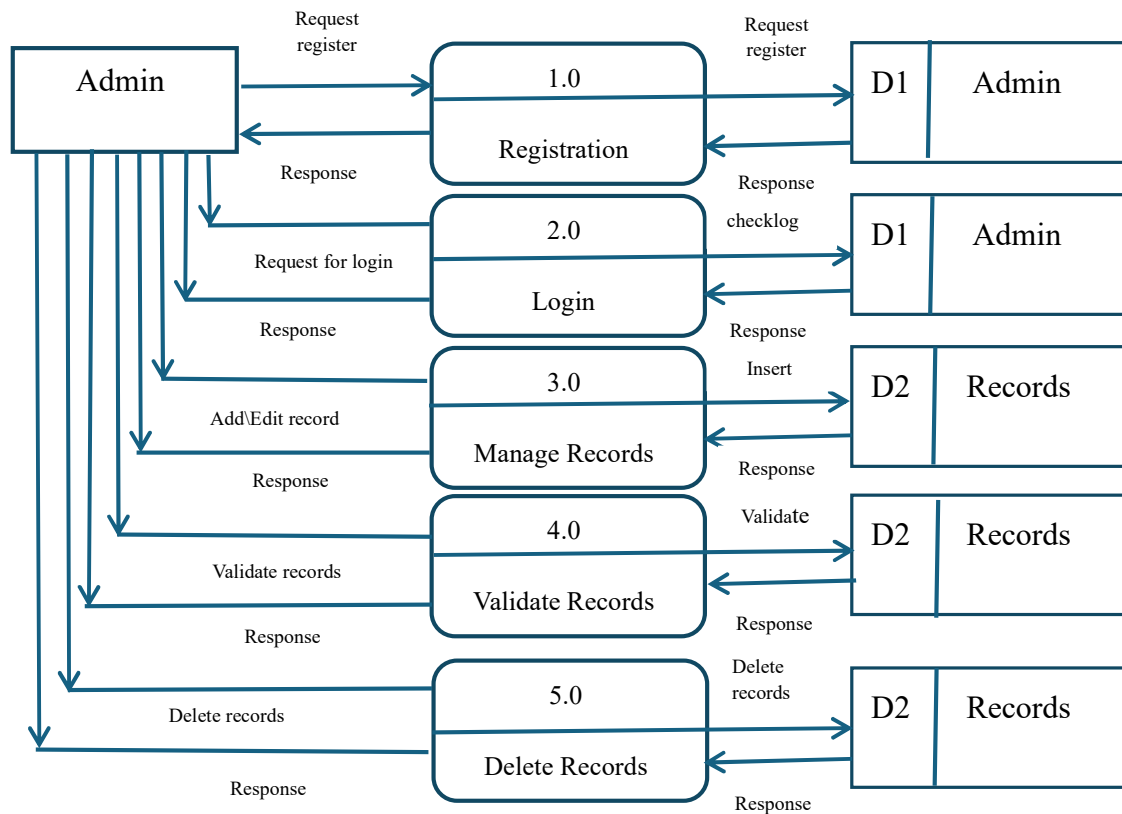


2.2 DATA FLOW DIAGRAM

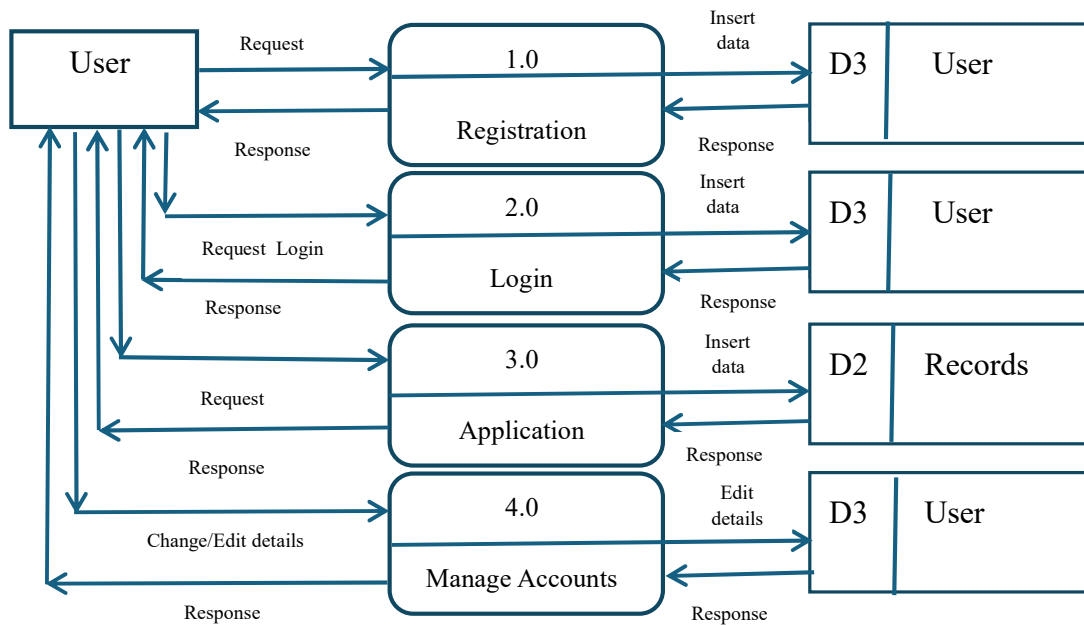
2.2.1 DFD LEVEL0



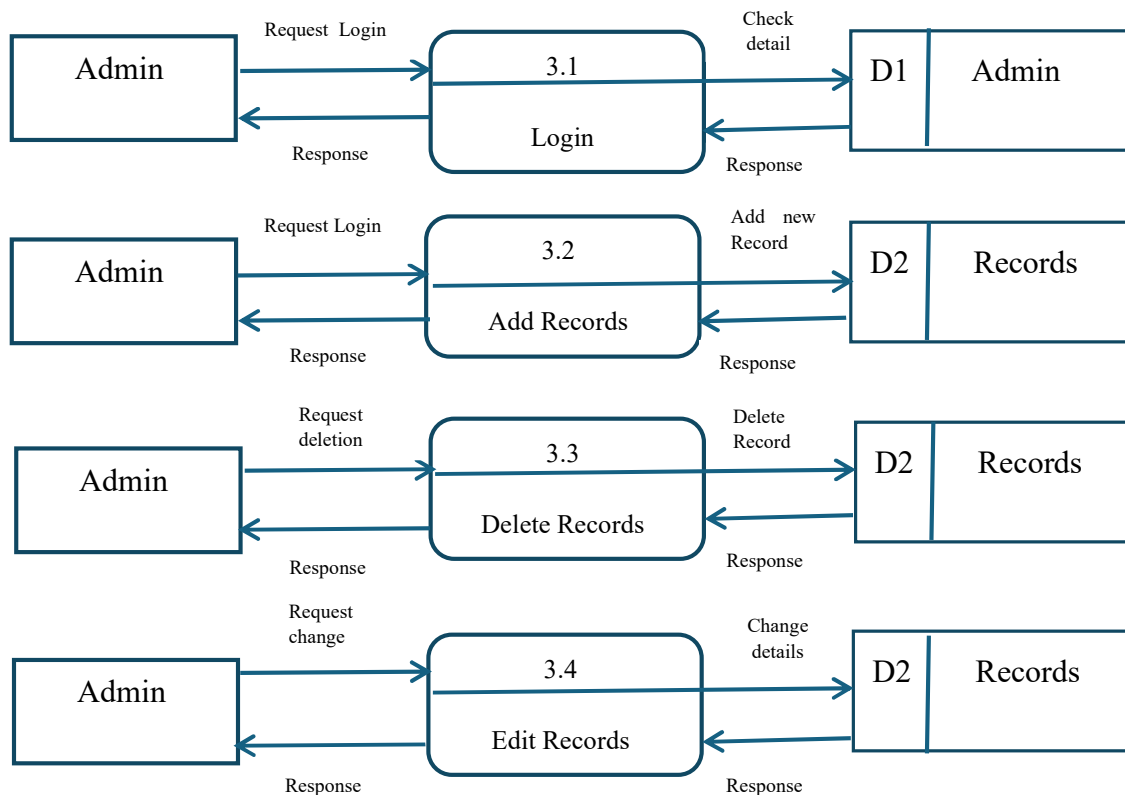
2.2.2 DFD LEVEL 1 : Admin



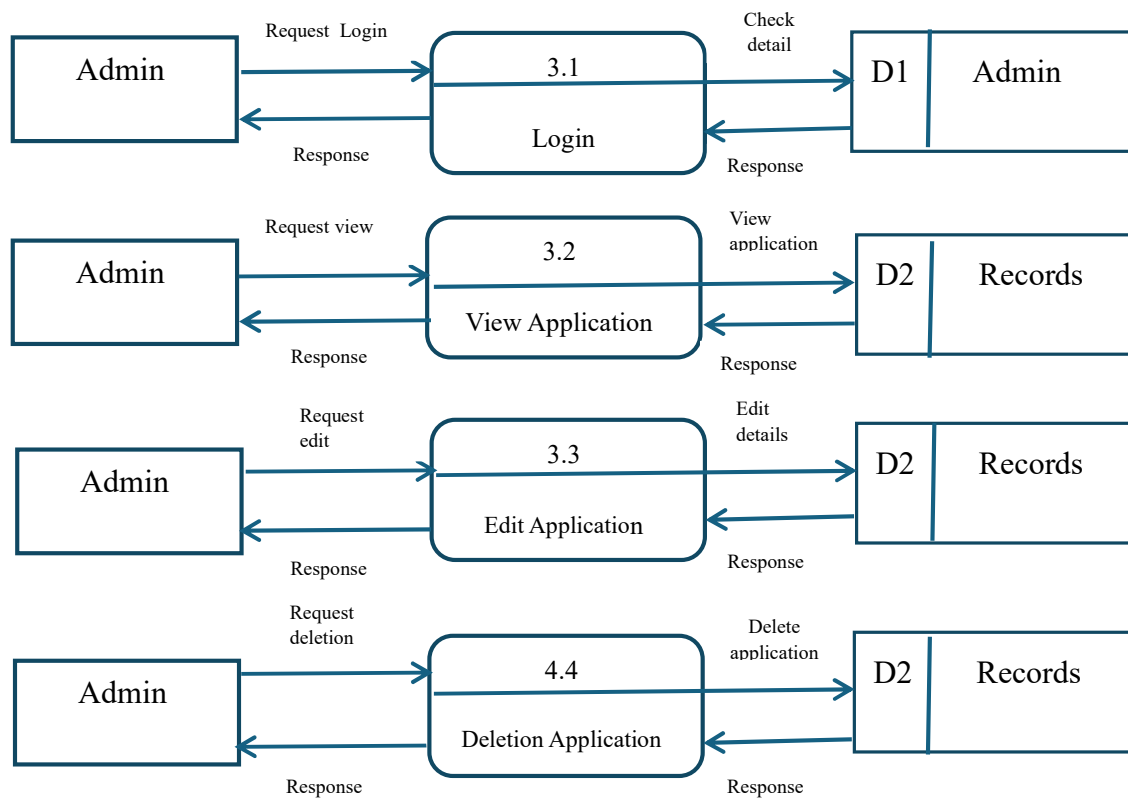
2.2.2 DFD LEVEL 1 : User



2.2.4 DFD LEVEL 2 : Admin



2.2.5 DFD LEVEL 2 : User



2.3 DATABASE DESIGN

The Hostel Management System uses a MySQL relational database to store and manage all system data efficiently. The database is designed using normalization techniques to reduce redundancy and maintain data integrity. Each module of the system is supported by separate tables that are interconnected using primary keys and foreign keys.

The database ensures secure storage of student information, room details, attendance records, complaints, food orders, and administrative data. Structured queries (SQL) are used to insert, update, delete, and retrieve data from the database.

2.3.1 TABLE NAME : admin

The name of the table is Admin. The table is used to store administrator details such as username, email, password, registration date and updation date.

S.NO	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	INT(11)	No	NONE	Admin ID
2	username	VARCHAR(255)	No	NONE	Login username
3	email	VARCHAR(255)	No	NONE	Admin email
4	password	VARCHAR(255)	No	NONE	Encrypted password
5	reg_date	TIMESTAMP	No	current_timestamp()	Registration date
6	updation_date	DATE	No	NONE	Last updated date

2.3.2 TABLE NAME : room_allotment

The name of the table is Room Allotment. The table is used to store room allocation details such as student ID, room ID, and allotment date.

S.NO	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	INT(11)	No	NONE	Allotment ID
2	student_id	INT(11)	Yes	NULL	Student reference ID
3	room_id	INT(11)	Yes	NULL	Room reference ID
4	allot_date	DATE	Yes	NULL	Allotment date

2.3.3 TABLE NAME : food

The name of the table is Food. The table is used to store food item details such as food name, price, and type.

S.NO	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	INT(11)	No	NONE	Food id
2	food_name	VARCHAR(255)	Yes	NULL	Food name
3	price	INT(11)	Yes	NULL	Food price
4	type	VARCHAR(255)	Yes	NULL	Food category
5	created_at	TIMESTAMP	No	current_timestamp()	Creation timestamp

2.3.4 TABLE NAME : complaints

The name of the table is Complaints. The table is used to store student complaint details such as complaint type, description, and status.

S.NO	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	INT(11)	No	NULL	Complaint ID
2	complainNumber	BIGINT(12)	Yes	NULL	Unique complaint number
3	userId	INT(11)	Yes	NULL	Student reference ID
4	complaintType	VARCHAR(255)	Yes	NULL	Type of complaint
5	complaintDetails	MEDIUMTEXT	Yes	NULL	Complaint description
6	complaintDoc	VARCHAR(255)	Yes	NULL	Supporting document
7	complaintStatus	VARCHAR(255)	Yes	NULL	Current status
8	registrationDate	TIMESTAMP	Yes	current_timestamp()	Complaint date

2.3.5 TABLE NAME : attendance

The name of the table is Attendance. The table is used to record student attendance details such as date, time, reason.

S.NO	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	INT(11)	No	NONE	Attendance ID
2	student_id	INT(11)	No	NONE	Student reference ID
3	attendance_date	DATE	No	NONE	Attendance date
4	attendance_time	TIME	No	NONE	Attendance time
5	status	ENUM('Present', Absent')	No	NONE	Present/Absent status
6	reason	VARCHAR(255)	Yes	NULL	Absent reason
7	Created_at	TIMESTAMP	No	current_timestamp()	Record creation time

2.3.6 TABLE NAME : feedback

The name of the table is Feedback. The table is used to store student ratings and feedback comments.

S.NO	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	INT(11)	No	NONE	Feedback ID
2	AccessibilityWarden	VARCHAR(255)	Yes	NULL	Warden accessibility rating
3	AccessibilityMember	VARCHAR(255)	Yes	NULL	Staff accessibility rating
4	RedressalProblem	VARCHAR(255)	Yes	NULL	Issue resolution rating
5	Room	VARCHAR(255)	Yes	NULL	Room condition rating
6	Mess	VARCHAR(255)	Yes	NULL	Mess quality rating
7	HostelSourroundings	VARCHAR(255)	Yes	NULL	Surrounding rating
8	OverallRating	VARCHAR(255)	Yes	NULL	Overall rating
9	FeedbackMessage	VARCHAR(255)	Yes	NULL	Feedback comment
10	userId	INT(11)	Yes	NULL	Student reference ID
11	PostingDate	TIMESTAMP	Yes	current_timestamp()	Submission date

2.3.7 TABLE NAME : courses

The name of the table is Courses. The table is used to store course details such as course code and course name.

S.NO	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	INT(11)	No	NONE	Course ID
2	course_code	VARCHAR(255)	Yes	NULL	Course code
3	course_sn	VARCHAR(255)	Yes	NULL	Course short name
4	course_fn	VARCHAR(255)	Yes	NULL	Course full name
5	posting_date	TIMESTAMP	Yes	current_timestamp()	Creation date

2.3.8 TABLE NAME : userregistration

The name of the table is User Registration. The table is used to store student account details such as name, email, password, and registration date.

S.NO	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	INT(11)	No	NONE	User ID
2	regNo	VARCHAR(255)	Yes	NULL	Registration number
3	firstName	VARCHAR(255)	Yes	NULL	First name
4	middleName	VARCHAR(255)	Yes	NULL	Middle name
5	lastName	VARCHAR(255)	Yes	NULL	Last name
6	gender	VARCHAR(255)	Yes	NULL	gender
7	contactNo	BIGINT(20)	Yes	NULL	Contact number
8	email	VARCHAR(255)	Yes	NULL	Email address
9	passsword	VARCHAR(255)	Yes	NULL	Account password
10	regDate	TIMESTAMP	Yes	current_timestamp()	Registration date
11	updaDate	VARCHAR(45)	Yes	NULL	Last updated date
12	passUdaDate	VARCHAR(45)	Yes	NULL	Password updaDate

2.3.9 TABLE NAME : student_fees

The name of the table is Student Fees. The table is used to store fee payment details such as student ID, amount, payment status, and paid date.

S.NO	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	INT(11)	No	NONE	Fee record ID
2	student_id	INT(11)	No	NONE	Student reference ID
3	room_no	VARCHAR(50)	No	NONE	Room number
4	fees	ENUM('Paid','Un paid')	No	NONE	Fee payment status
5	status	INT(11)	No	NONE	Payment status code
6	paid_date	DATE	Yes	NULL	Fee payment date
7	created_at	TIMESTAMP	No	current_timestamp()	Record creation date

2.3.10 TABLE NAME : rooms

The name of the table is Rooms. The table is used to store room details such as room_no and fee.

S.NO	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	INT(11)	No	NONE	Room ID
2	seater	INT(11)	Yes	NULL	Seating capacity
3	room_no	INT(11)	Yes	NULL	Room number
4	fees	INT(11)	Yes	NULL	Room fees
5	posting_date	timestamp	Yes	time_stamp()	Creation date

2.3.11 TABLE NAME : registration

The name of the table is Registration. The table is used to store student personal, academic, and contact details.

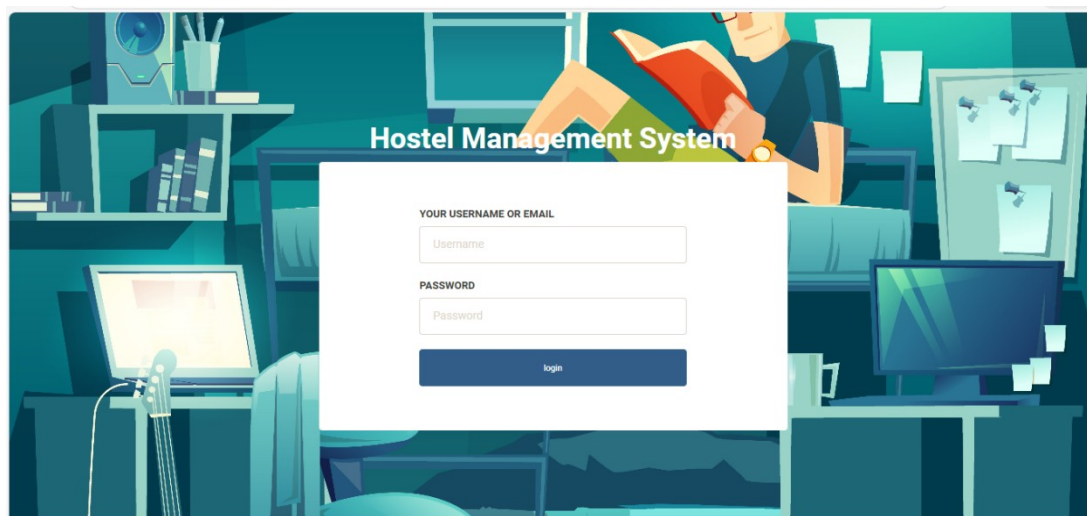
S.NO	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	INT(11)	No	NONE	Registration ID
2	roomno	INT(11)	Yes	NULL	Room number
3	seater	INT(11)	Yes	NULL	Number of seats
4	feespm	INT(11)	Yes	NULL	Fees per month
5	foodstatus	INT(11)	Yes	NULL	Food faculty status
6	roomfrom	DATE	Yes	NULL	Room allocation date
7	duration	INT(11)	Yes	NULL	Stsy duration
8	course	VARCHAR(500)	Yes	NULL	Course name
9	regno	INT(11)	Yes	NULL	Registration number
10	firstName	VARCHAR(500)	Yes	NULL	Student first name
11	middleName	VARCHAR(500)	Yes	NULL	Student middle name
12	lastName	VARCHAR(500)	Yes	NULL	Student last name
13	gender	VARCHAR(500)	Yes	NULL	gender
14	contactno	BIGINT(11)	Yes	NULL	Student contact number
15	emailid	VARCHAR(500)	Yes	NULL	Student email
16	egycontactno	BIGINT(11)	Yes	NULL	Emergency contact number
17	guardianName	VARCHAR(500)	Yes	NULL	Guardian name
18	guardianRelation	VARCHAR(500)	Yes	NULL	Guardian relation

19	guardianContactno	BIGINT(11)	Yes	NULL	Guardian contact number
20	corresAddress	VARCHAR(500)	Yes	NULL	Correspondance address
21	corresCity	VARCHAR(500)	Yes	NULL	Correspondance city
22	corresState	VARCHAR(500)	Yes	NULL	Correspondance state
23	corresPincode	INT(11)	Yes	NULL	Correspondance pincode
24	pmntAddress	VARCHAR(500)	Yes	NULL	Permanenet address
25	pmntCity	VARCHAR(500)	Yes	NULL	Permanent city
26	pmnatetState	VARCHAR(500)	Yes	NULL	Permananet state
27	pmntPincode	INT(11)	Yes	NULL	Permaneent pincode
28	postingDate	TIMESTAMP	Yes	current_timest amp()	Registration date
29	updationDate	VARCHAR(500)	Yes	NULL	Last updated date

2.4 FORM DESIGN

2.4.1 Admin Login Page:

This page is used by the admin to log in using username or email and password. It gives secure access to the hostel management system.



2.4.2 Add Course Page:

This page is used by the admin to add new course details. The admin can enter course code and course name and save it in the system.

The screenshot shows the 'Add Courses' page within the 'Hostel Management System'. The page has a dark sidebar on the left with a 'MAIN' menu containing items like Dashboard, Courses, Rooms, Attendance, Fees, Canteen, Student Registration, Manage Students, Complaints, Feedback, and User Access logs. The top header displays 'Hostel Management System' and an 'Account' dropdown. The main content area is titled 'Add Courses' and features a form with three input fields: 'Course Code', 'Course Name (Short)', and 'Course Name (Full)'. A blue 'Add course' button is positioned below the 'Course Name (Full)' field.

2.4.3 Add Room Page:

This page is used by the admin to add new room details. The admin can enter room number, seating capacity, and fee per student and save it in the system.

The screenshot shows the 'Add a Room' page within the 'Hostel Management System'. The layout is consistent with the previous page, featuring a dark sidebar and a top header. The main content area is titled 'Add a Room' and contains a form with three input fields: 'Select Seater' (a dropdown menu), 'Room No.', and 'Fee(Per Student)'. A blue 'Create Room' button is located below the 'Fee(Per Student)' field.

2.4.4 Mark Attendance Page:

This page is used by the admin to mark student attendance. The admin can select present or absent status and save the attendance record.

The screenshot shows the 'Mark Attendance' page within the 'Hostel Management System'. The page has a dark sidebar with a 'MAIN' menu containing links to Dashboard, Courses, Rooms, Attendance, Fees, Canteen, Student Registration, Manage Students, Complaints, Feedback, and User Access logs. The 'Attendance' link is highlighted. The main content area is titled 'Mark Attendance' and contains an 'ATTENDANCE SHEET' section. At the top of this section is a date input field with the placeholder 'dd-mm-yyyy'. Below it is a table with four columns: 'S No', 'Student Name', 'Status', and 'Absent Reason'. The table contains eight rows of data, all with 'Present' in the Status column. A 'Save Attendance' button is located at the bottom left of the table.

S No	Student Name	Status	Absent Reason
1	Anuj kumar	Present	
2	John Doe	Present	
3	Anuj kumar	Present	
4	John Matthew	Present	
5	Amit Singh	Present	
6	mithya s	Present	
7	Raj N	Present	
8	Meena k	Present	

2.4.5 Add Food Page:

This page is used by the admin to add new food items to the system. The admin can enter food name, price, and type and save it.

The screenshot shows the 'Add Food' page within the 'Hostel Management System'. The sidebar is identical to the previous page, with the 'Attendance' link highlighted. The main content area is titled 'Add Food' and contains an 'ADD FOOD ITEM' section. This section has three input fields: 'Food Name', 'Price (₹)', and 'Food Type' (a dropdown menu with 'Select Type' as the current selection). A blue 'Add Food' button is positioned below the 'Food Type' dropdown.

2.4.6 Student Fees Entry Page:

This page is used by the admin to enter and manage student fee details. The admin can select a student, enter fee amount, status, and paid date, and save the record.

The screenshot shows the 'Student Fees Entry' page within the 'Hostel Management System'. The page has a dark sidebar on the left with a 'MAIN' menu containing: Dashboard, Courses, Rooms, Attendance, Fees, Canteen, Student Registration, Manage Students, Complaints, Feedback, and User Access logs. The main content area is titled 'Student Fees Entry' and contains a form labeled 'ADD STUDENT FEES'. The form fields are: 'Student' (a dropdown menu with 'Select Student' as the placeholder), 'Register No' (a text input field), 'Room No' (a text input field), 'Fees' (a text input field), 'Status' (a dropdown menu with 'Paid' as the selected option), and 'Paid Date' (a date picker showing 'dd-mm-yyyy'). A blue 'Save Fees' button is located at the bottom of the form. In the top right corner of the system header, there is an 'Account' link with a user profile icon.

2.4.7 User Login Page:

This page is used by users to log in using their email or registration number and password. It provides secure access to the hostel management system.

The screenshot shows the 'User Login' page within the 'Hostel Management System'. The page has a dark sidebar on the left with a 'MAIN' menu containing: User Registration, User Login, and Admin Login. The main content area is titled 'User Login' and contains a login form. The form has two input fields: 'EMAIL / REGISTRATION NUMBER' (with placeholder text 'Email / Registration Number') and 'PASSWORD' (with placeholder text 'Password'). Below these fields is a blue 'login' button. At the bottom of the form, there is a link that says 'Forgot password?'. The system header at the top left displays 'Hostel Management System'.

2.4.8 Register Complaint Page:

This page is used by students to submit complaints about hostel issues. They can select the complaint type, explain the problem, and upload a file if needed.

The screenshot shows the 'Register Complaint' page within the 'Hostel Management System'. The page has a dark sidebar on the left with a 'MAIN' menu containing: Dashboard, Book Hostel, Room Details, Attendance, Canteen, Complaint Registration, Registered Complaints, Feedback, My Profile, Change Password, and Access log. The top header shows 'Hostel Management System' and an 'Account' dropdown. The main content area is titled 'Register Complaint' and contains a form with three fields: 'Complaint Type' (a dropdown menu with 'Select Complaint Type'), 'Explain the Complaint' (a text area), and 'File (if any)' (a file upload button labeled 'Choose File' and the text 'No file chosen'). A blue 'Submit' button is located at the bottom right of the form.

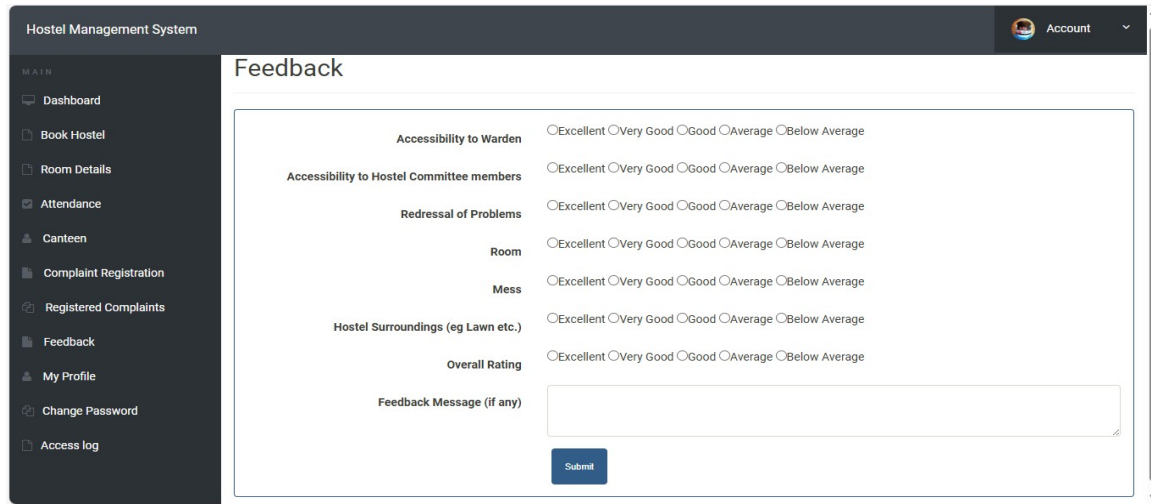
2.4.9 Order Food Page:

This page allows students to select food items, enter quantity, and place food orders online. The order details are stored in the system for admin monitoring.

The screenshot shows the 'Order Food' page within the 'Hostel Management System'. The layout is identical to the previous page, with the same sidebar and header. The main content area is titled 'Order Food' and contains a form with three fields: 'Student' (a dropdown menu with 'Select Student'), 'Food Item' (a dropdown menu with 'Select Food'), and 'Quantity' (a text input field). A blue 'Place Order' button is located at the bottom right of the form.

2.4.10 Feedback Page:

This page enables students to give ratings and feedback about hostel facilities and services. It helps the administration improve hostel management and student satisfaction.



The screenshot shows the 'Feedback' page within the 'Hostel Management System'. The page has a dark sidebar on the left with a 'MAIN' menu containing: Dashboard, Book Hostel, Room Details, Attendance, Canteen, Complaint Registration, Registered Complaints, Feedback (highlighted), My Profile, Change Password, and Access log. The main content area is titled 'Feedback' and contains a form with the following sections:

- Accessibility to Warden**: Radio buttons for Excellent, Very Good, Good, Average, Below Average.
- Accessibility to Hostel Committee members**: Radio buttons for Excellent, Very Good, Good, Average, Below Average.
- Redressal of Problems**: Radio buttons for Excellent, Very Good, Good, Average, Below Average.
- Room**: Radio buttons for Excellent, Very Good, Good, Average, Below Average.
- Mess**: Radio buttons for Excellent, Very Good, Good, Average, Below Average.
- Hostel Surroundings (eg Lawn etc.)**: Radio buttons for Excellent, Very Good, Good, Average, Below Average.
- Overall Rating**: Radio buttons for Excellent, Very Good, Good, Average, Below Average.
- Feedback Message (if any)**: A text input field.
- Submit**: A blue button at the bottom right of the form.

2.5 MODULE EXPLANATION

The system is designed as a modular web-based Hostel Management System that integrates student management, room allocation, fee tracking, attendance monitoring, food management, and complaint handling into a single platform. Each module performs a specific function and interacts with the central database to ensure data consistency and smooth system operation. The Admin module manages courses, rooms, students, fees, attendance, food items, complaints, and report generation, while the Student module handles registration, room booking, fee payment, food ordering, attendance viewing, complaint submission, and profile management. The system uses secure authentication and session management mechanisms for both admin and student access to ensure data privacy and protection. All records are stored in the database and displayed through a dashboard interface for easy monitoring and control.

The system is divided into two main modules:

- Admin Module
- Student Module

2.5.1 Admin Module

The Admin module manages overall system operations, including student management, room allocation, attendance tracking, fee management, food management, complaint resolution, and report generation.

S. NO	MODULE NAME	EXPLANATION
1.	Admin Login	Admin can securely log in to the system using valid credentials.
2.	Admin Profile	Admin can view and update personal profile details and change the password.
3.	Courses	Admin can add, edit, and delete course details in the system.
4.	Rooms	Admin can create rooms, assign seaters, set fees, and manage room allocation.
5.	Registration	Admin can register students and allocate rooms based on availability.
6.	Canteen	Admin can manage food items, update prices, and generate food reports.
7.	Attendance	Admin can mark, manage, and monitor student attendance records.
8.	Complaints	Admin can view, update, and manage student complaints with status tracking.
9.	Feedback	Admin can view and analyze feedback submitted by students.
10.	Forgot Password	Admin can reset the password using the forgot password option.
11.	Admin Access Log	Admin can monitor login history and access details for security purposes.

2.5.2 Student Module

The Student module allows students to register, book rooms, pay fees, view attendance, order food, submit complaints, and manage their personal profiles.

S. NO	MODULE NAME	EXPLANATION
1.	User Registration	User can register in the system by filling the online registration form with required details.
2.	User Login	User can log in securely using registered credentials.
3.	User Profile	User can view and update personal profile information.
4.	Book Hostel	User can check room availability and book a hostel room online.
5.	View Attendance	User can view personal attendance records maintained by the admin.
6.	Order Food	User can order food items from the hostel canteen through the system.
7.	View Food Order	User can view the history and details of ordered food items.
8.	Room details	User can view details of the allotted room including room number and fees.
9.	Forgot Password	User can reset the password using the forgot password option.
10.	User Access Log	User can view last login details for security purposes.

CHAPTER 3

SYSTEM IMPLEMENTATION

3.1 IMPLEMENTATION DETAILS FOR EACH MODULE

The Hostel Management System is implemented as a web-based application to automate hostel administration activities. The system is developed using PHP as the server-side programming language, MySQL as the database, and HTML, CSS, and JavaScript for the user interface.

The system follows a modular architecture where different modules handle specific tasks of hostel management. In this project, the two main modules are Admin Module and Student Module. Each module interacts with the central database to store, retrieve, and update hostel data securely. The modular implementation improves system efficiency, maintainability, and scalability.

3.1.1 Admin Module

The Admin Module allows the administrator to manage all hostel activities through the system. After logging in, the admin can access the dashboard and manage student details, room allocation, attendance records, food orders, complaints, and feedback. The admin can add, update, or delete student and room information. Complaints submitted by students can also be viewed and updated by the admin. All data is processed using PHP and stored in the MySQL database, which helps the admin manage hostel operations efficiently.

START

ADMIN enters username and password

IF credentials are valid THEN

DISPLAY Admin Dashboard

ELSE

DISPLAY "Invalid Login"

END IF

WHILE Admin is logged in DO

DISPLAY options:

1. Manage Students
2. Manage Rooms
3. View Bookings
4. View Complaints
5. View Food Orders
6. Generate Reports
7. Logout

ADMIN selects option

IF option = Manage Students THEN

ADD / UPDATE / DELETE student details

SAVE changes in database

END IF

IF option = Manage Rooms THEN

 ADD room details

 UPDATE room availability

 STORE room data in database

END IF

IF option = View Bookings THEN

 RETRIEVE booking data from database

 DISPLAY booking details

END IF

IF option = View Complaints THEN

 RETRIEVE complaints from database

 UPDATE complaint status

END IF

IF option = View Food Orders THEN

 RETRIEVE food orders

UPDATE order status

```
END IF
```

```
IF option = Generate Reports THEN
```

```
    FETCH required data
```

```
    CREATE report
```

```
    DISPLAY or DOWNLOAD report
```

```
END IF
```

```
IF option = Logout THEN
```

```
    EXIT system
```

```
END IF
```

```
END WHILE
```

```
STOP
```

3.1.2 Student Module

The Student Module allows students to use hostel services through the system. Students can register and log in securely to access their accounts. After login, they can view their profile, check room availability, request room booking, view attendance, order food, submit complaints, and give feedback. All the information entered by students is stored in the database and can be managed by the admin.

START

STUDENT enters username and password

IF credentials are valid THEN

DISPLAY Student Dashboard

ELSE

 DISPLAY "Invalid Login"

END IF

WHILE Student is logged in DO

 DISPLAY options:

 1. View Profile

 2. Book Room

 3. Check Attendance

 4. Order Food

 5. Submit Complaint

 6. Give Feedback

 7. Logout

STUDENT selects option

IF option = View Profile THEN

 FETCH student details from database

 DISPLAY profile

END IF

IF option = Book Room THEN

 CHECK available rooms

 SELECT room

 SAVE booking in database

END IF

IF option = Check Attendance THEN

 RETRIEVE attendance data

 DISPLAY attendance

END IF

IF option = Order Food THEN

 SELECT food items

 PLACE order

STORE order in database

END IF

IF option = Submit Complaint THEN

ENTER complaint details

SAVE complaint in database

END IF

IF option = Give Feedback THEN

ENTER feedback

SAVE feedback in database

END IF

IF option = Logout THEN

EXIT system

END IF

END WHILE

STOP

CHAPTER 4

SYSTEM TESTING

4.1 Unit Testing

Unit Testing was performed to test each module of the Hostel Management System individually. Every module was tested separately to ensure that it works correctly without depending on other modules.

The Admin Login module was tested to verify correct username and password validation. The Student Registration module was tested to ensure proper data storage in the database. The Add Room module was tested for correct room entry and availability status. Attendance, Fees, Food Management, and Complaint modules were also tested individually to check input validation, data insertion, update, and deletion operations. This testing helped to identify errors at the early stage and ensured that each module functions correctly.

4.2 Integration Testing

Integration Testing was conducted after combining all modules together. This testing ensured that different modules interact properly with each other and share data correctly.

For example, after student registration, the details were verified in the Admin panel. After room allocation, the assigned room was checked in the student dashboard. Food orders placed by students were verified in the Admin food management section. Complaints submitted by students were checked in the Admin complaint module. This testing confirmed smooth data flow between modules and proper database connectivity.

4.3 System Testing

System Testing was performed to evaluate the complete Hostel Management System as a whole. All modules such as Admin, Student, Room Management, Attendance, Fees, Food, and Complaints were tested together.

The system was tested for correct login authentication, proper navigation between pages, accurate data display, report generation, and dashboard updates. All functionalities were verified to ensure that the system meets the project requirements. The testing confirmed that the overall system works efficiently without major errors.

4.4 User Acceptance Testing (UAT)

User Acceptance Testing was conducted to verify whether the system meets the real-time requirements of hostel management. This testing was performed by end users (Admin and Students).

The Admin tested student management, room allocation, attendance tracking, fee management, food management, and complaint handling features. Students tested registration, login, room booking, food ordering, fee checking, complaint submission, and profile management. Users confirmed that the system is user-friendly, easy to operate, and satisfies their requirements. No major issues were reported during this phase.

CHAPTER 5

FUTURE ENHANCEMENT

The current Hostel Management System effectively automates major hostel activities such as student registration, room allocation, attendance management, food ordering, complaint handling, and administrative monitoring. While the system meets the present requirements, it can be further enhanced with additional advanced features to improve efficiency, security, and scalability.

In the future, an online payment gateway can be integrated to allow students to pay hostel fees and food charges through secure digital payment methods such as debit cards, credit cards, UPI, and net banking. A mobile application version for Android and iOS platforms can also be developed to provide easier and faster access to hostel services. Biometric or RFID-based attendance systems can be implemented to ensure accurate and real-time attendance tracking.

The system can also be improved by adding SMS and email notification features to send alerts for fee reminders, complaint status updates, and room allocation confirmations. Advanced graphical reports and data analytics can help administrators make better decisions. Cloud deployment and enhanced role-based access control can further improve data security, scalability, and system reliability. With these enhancements, the Hostel Management System can become a more powerful, secure, and fully automated solution for modern hostel management.

CHAPTER 6

CONCLUSION

The Hostel Management System has been successfully designed and developed to automate and simplify the management of hostel operations. The system effectively replaces manual record-keeping methods with a secure and centralized web-based platform. It provides separate modules for administrators and students, ensuring structured management and controlled access to data.

Through this system, major hostel activities such as student registration, room allocation, attendance management, food ordering, complaint handling, and feedback management are efficiently handled. The Admin module enables effective monitoring and control of hostel activities, while the Student module provides convenient access to essential services such as room booking, attendance viewing, and complaint submission. The use of PHP and MySQL ensures reliable backend processing and secure data storage. Proper authentication, session management, and database design contribute to maintaining data integrity and system security. The system reduces paperwork, minimizes human errors, improves transparency, and saves time for both administrators and students.

Overall, the Hostel Management System provides an organized, efficient, and scalable solution for managing hostel operations. The project demonstrates how web technologies can be effectively utilized to modernize traditional administrative processes and enhance overall operational efficiency.

CHAPTER 7

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CHAPTER 8

APPENDIX I

HARDWARE REQUIREMENTS

The Hostel Management System does not require high-end hardware. The following minimum hardware configuration is sufficient:

- **Processor** : Intel Core i3 or above
- **RAM** : 4 GB or above
- **Hard Disk** : 500 GB or above
- **Monitor** : 15 inch colour display
- **Keyboard** : Standard Keyboard

APPENDIX II

SOFTWARE REQUIREMENTS

The system is developed as a web-based application using the following software technologies:

- **Operating System** : Windows 11
- **Front-End** : HTML, CSS, JavaScript
- **Back-End** : PHP
- **Database** : MySQL
- **Server** : XAMPP Server
- **Web Browser** : Microsoft Edge

APPENDIX III

SAMPLE SCREENSHOTS

User Sign In

The screenshot shows the 'User Login' page of the 'Hostel Management System'. On the left is a dark sidebar with a 'MAIN' menu containing 'User Registration', 'User Login', and 'Admin Login'. The main content area has a title 'User Login' and a login form. The form includes two input fields: 'EMAIL / REGISTRATION NUMBER' and 'PASSWORD'. Below these is a blue 'login' button and a link for 'Forgot password?'.

Hostel Management System

MAIN

- User Registration
- User Login
- Admin Login

User Login

EMAIL / REGISTRATION NUMBER

PASSWORD

login

[Forgot password?](#)

User Dashboard

The screenshot shows the 'Dashboard' page of the 'Hostel Management System'. The top header includes the system name and a user account dropdown. The left sidebar lists various dashboard options. The main content area features two large cards: 'My Profile' and 'My Room', each with a 'FULL DETAIL' or 'SEE ALL' link.

Hostel Management System

Account

MAIN

- Dashboard
- Book Hostel
- Room Details
- Attendance
- Canteen
- Complaint Registration
- Registered Complaints
- Feedback
- My Profile
- Change Password
- Access log

Dashboard

My Profile

FULL DETAIL →

My Room

SEE ALL →

User Booked Hostel Details

Hostel Management System

Account

MAIN

Dashboard

Book Hostel

Room Details

Attendance

Canteen

Complaint Registration

Registered Complaints

Feedback

My Profile

Change Password

Access log

Rooms Details

ALL ROOM DETAILS

Room Related Info

Registration Number :	15	Apply Date :	2026-01-29 11:25:56		
Room no :	132	Seater :	5	Fees PM :	2000
Food Status:	With Food	Stay From :	2026-01-29	Duration:	2 Months
Hostel Fee:	4000	Food Fee:	4000		
Total Fee :	8000				

Personal Info

Reg No. :	15	Full Name :	Meenak	Email :	meena@gmail.com
Contact No. :	9788879909	Gender :	female	Course :	Bachelor of Science
Emergency Contact No. :	9788879909	Guardian Name :	-	Guardian Relation :	-
Guardian Contact No. :	0				

Addresses

Correspondence Address	no : 51 ,muthumariyaman street,meenakshipuram karakudi, 630002 Tamil Nadu	Permanent Address	no : 51 ,muthumariyaman street,meenakshipuram karakudi, 630002 Tamil Nadu		
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User View Attendance

Hostel Management System

Account

MAIN

Dashboard

Book Hostel

Room Details

Attendance

Canteen

Complaint Registration

Registered Complaints

Feedback

My Profile

Change Password

Access log

View Attendance

SELECT DATE

dd-mm-yyyy

View Attendance

ATTENDANCE LIST (2026-01-14)

S.No	Student Name	Status
1	Anuj kumar	Present
2	John Doe	Absent
3	Anuj kumar	Present
4	John Matthew	Absent
5	Amit Singh	Present
6	nithya s	Present

User Order Food

Hostel Management System

Account

MAIN

Dashboard

Book Hostel

Room Details

Attendance

Canteen

Complaint Registration

Registered Complaints

Feedback

My Profile

Change Password

Access log

Order Food

FOOD ORDER

Student

John Matthew

Food Item

Curd Rice - ₹60

Quantity

1

Place Order

User Register Complaint

Hostel Management System

Account

MAIN

Dashboard

Book Hostel

Room Details

Attendance

Canteen

Complaint Registration

Registered Complaints

Feedback

My Profile

Change Password

Access log

Register Complaint

Complaint Type

Select Complaint Type

Explain the Complaint

File (if any)

Choose File

No file chosen

Submit

User Feedback

Hostel Management System

 Account

MAIN

Dashboard

Book Hostel

Room Details

Attendance

Canteen

Complaint Registration

Registered Complaints

Feedback

My Profile

Change Password

Access log

Feedback

Accessibility to Warden

Accessibility to Hostel Committee members

Redressal of Problems

Room

Mess

Hostel Surroundings (eg Lawn etc.)

Overall Rating

Feedback Message (if any)

☐Excellent

☐Very Good

☐Good

☐Average

☐Below Average

☐Excellent

☐Very Good

☐Good

☐Average

☐Below Average

☐Excellent

☐Very Good

☐Good

☐Average

☐Below Average

☐Excellent

☐Very Good

☐Good

☐Average

☐Below Average

☐Excellent

☐Very Good

☐Good

☐Average

☐Below Average

☐Excellent☐Very Good☐Good☐Average☐Below Average☐Excellent☐Very Good☐Good☐Average☐Below Average

Submit

Admin Login

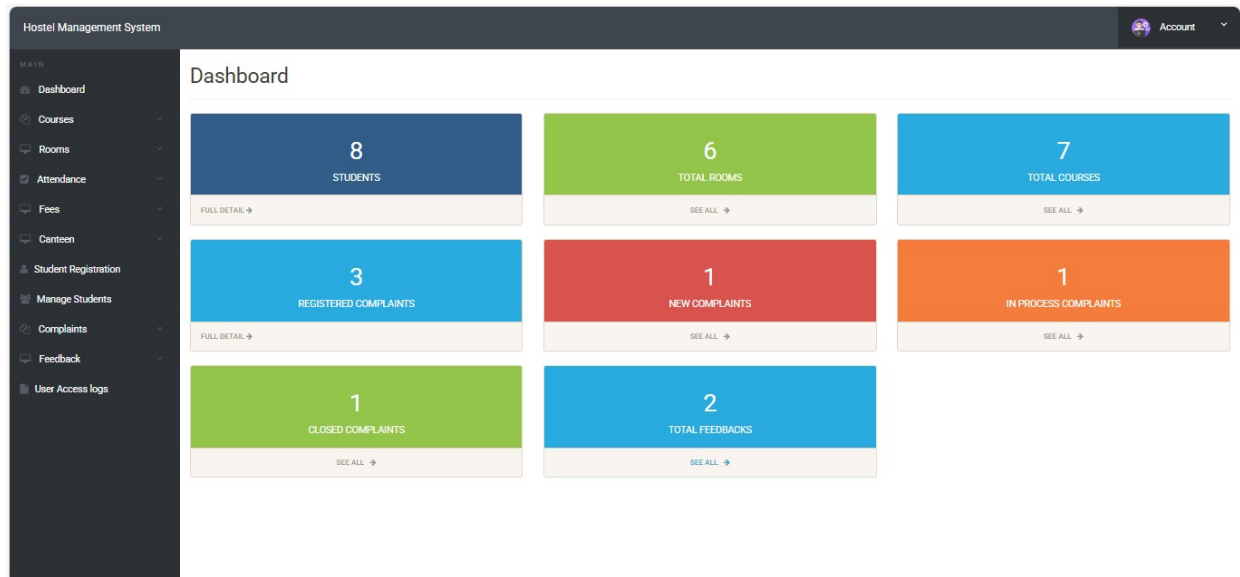
Hostel Management System

YOUR USERNAME OR EMAIL

PASSWORD

login

Admin Dashboard



Admin Manage Room

The Admin Manage Room page displays a table of room details. The table has 7 columns: Sno., Seater, Room No., Fees (PM), Posting Date, and Action. The table shows 6 entries.

Sno.	Seater	Room No.	Fees (PM)	Posting Date	Action
1	5	100	8000	2025-01-02 04:15:43	Edit Delete
2	2	200	6000	2025-01-02 04:15:43	Edit Delete
3	3	112	4000	2025-01-02 04:15:43	Edit Delete
4	5	132	2000	2025-01-02 04:15:43	Edit Delete
5	1	800	8000	2026-01-12 21:40:03	Edit Delete
6	3	301	1	2026-02-14 19:55:23	Edit Delete

Showing 1 to 6 of 6 entries

Navigation: PREVIOUS 1 NEXT

Manage Registered Students

Hostel Management System

Account

MAIN

Dashboard

Courses

Rooms

Attendance

Fees

Canteen

Student Registration

Manage Students

Complaints

Feedback

User Access logs

Manage Registerd Students

ALL ROOM DETAILS

Show 10 entries Search:

Sno.	Student Name	Reg no	Contact no	room no	Seater	Staying From	Action
1	Anujkumar	10806121	1234567890	100	5	2025-01-06	
2	JohnDoe	108061233	1425362514	200	2	2025-01-10	
3	Anujkumar	10806121	1234567890	200	2	2025-01-08	
4	JohnMatthew	14563213	8956237845	100	5	2025-01-15	
5	AmitkumarSingh	14563213	9632587412	132	5	2025-01-15	
6	nithyashris	18	8072194295	132	5	2026-01-12	
7	RajN	19	9988776655	132	5	2026-01-24	
8	Meenak	15	9788879909	132	5	2026-01-29	

Showing 1 to 8 of 8 entries

PREVIOUS 1 NEXT

Admin Mark Attendance

Hostel Management System

Account

MAIN

Dashboard

Courses

Rooms

Attendance

Fees

Canteen

Student Registration

Manage Students

Complaints

Feedback

User Access logs

ATTENDANCE SHEET

22-02-2026

S.No	Student Name	Status	Absent Reason
1	Anuj kumar	Present	
2	John Doe	Absent	going to sister marriage
3	Anuj kumar	Present	
4	John Matthew	Present	
5	Arnit Singh	Present	
6	nithya s	Present	
7	Raj N	Absent	suffering from fever
8	Meena k	Present	

Save Attendance

Admin Enter Student fees

Hostel Management System

Account

MAIN

Dashboard

Courses

Rooms

Attendance

Fees

Canteen

Student Registration

Manage Students

Complaints

Feedback

User Access logs

Student Fees Entry

ADD STUDENT FEES

Student

nithya s

Register No

18

Room No

132

Fees

2000

Status

Paid

Paid Date

22-02-2026

Save fees

Admin View Monthly Food Report

Hostel Management System

Account

MAIN

Dashboard

Courses

Rooms

Attendance

Fees

Canteen

Student Registration

Manage Students

Complaints

Feedback

User Access logs

Food Orders

VIEW FOOD ORDERS

Show 10 entries Search:

#	Student Name	Food Item	Price	Quantity	Total Price (₹)	Order Date	Status
1	Meena k	Mutton Biryani	₹300	1	₹300	2026-01-29	Ordered
2	Anuj kumar	Pongal	₹50	2	₹100	2026-01-29	Ordered
3	John Matthew	Chapati	₹40	2	₹80	2026-01-28	Ordered
4	Raj N	Egg Noodles	₹120	3	₹360	2026-01-27	Ordered
5	nithya s	curd rice	₹25	2	₹50	2026-01-27	Ordered

Showing 1 to 5 of 5 entries

PREVIOUS1NEXT

Admin View Complaints

Hostel Management System

 Account

MAIN

Dashboard

Courses

Rooms

Attendance

Fees

Canteen

Student Registration

Manage Students

Complaints

Feedback

User Access logs

New Complaints

COMPLAINT DETAILS

Show 10 entries Search:

Sno.	Complaint Number	Complaint Type	Complaint Status	Complaint Reg. Date	Action
1	731746111	Room Related	New	2026-01-29 11:20:50	

Showing 1 to 1 of 1 entries

PREVIOUS 1 NEXT

Admin View Feedback

Hostel Management System

 Account

MAIN

Dashboard

Courses

Rooms

Attendance

Fees

Canteen

Student Registration

Manage Students

Complaints

Feedback

User Access logs

Student Feedback

ALL ROOM DETAILS

Show 10 entries Search:

Sno.	Student Name	Reg no	room no	Seater	Staying From	Feedback Date	Action
1	JohnMatthew	14563213	100	5	2025-01-15	2025-01-14 16:42:43	
2	nithyashris	18	132	5	2026-01-12	2026-01-21 12:12:01	

Showing 1 to 2 of 2 entries

PREVIOUS 1 NEXT